INCH-POUND

MIL-PRF-32294 (AR)
13 June 2008

PERFORMANCE SPECIFICATION

CAP, BLASTING, NON-ELECTRIC - M7
CAP, BLASTING, PRACTICE, INERT, NON-ELECTRIC - XM30

This specification is approved for use by the U.S. Army Armament Research, Development, and Engineering Center (ARDEC) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the performance requirements and the verification provisions for the CAP, BLASTING: non-electric, M7 and the CAP, BLASTING, PRACTICE: inert, non-electric, M7.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are in sections 3 or 4 of this specification. This section may not include all documents referred to in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all the specified requirements of documents cited in sections 3 and 4 of this specification, whether or not documents are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

NORTH ATLANTIC TREATY ORGANIZATION STANDARDIZATION – NATO

AOP-7 Manual of Data Requirements and Tests for the Qualification of Explosive Materials for Military Use

Comments, suggestions, or questions on this document should be addressed to Commander, ARDEC, ATTN: AMSRD-AAR-QES-E., Picatinny, NJ 07806-5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil

AMSC N/A FSC 1375

DISTRIBUTION STATEMENT A: Approved for public release distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing
MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-464 - Department of Defense Interface Standard, Electromagnetic Environmental Effects, Requirements for Systems
MIL-STD-650 - Explosive: Sampling, Inspection and Testing
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-1168 - Ammunition Lot Numbering
MIL-STD-1916 - DOD Preferred Methods for Acceptance of Products
MIL-STD-2105 - Hazard Assessment Tests for Non-Nuclear Munitions

(These documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Documents Order Desk, 700 Robbins Avenue Bldg. 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those specified in the solicitation or contract (see 6.2).

PUBLICATIONS

TB 700-2 – Department of Defense Explosives Hazard Classification

(This document is available online at https://www3.dac.army.mil/es/documents/TB700_2.pdf, or from the DoD Explosives Safety Board, Room 856C, Hoffman Building 1, 2461 Eisenhower Avenue, Alexandria, VA 22331-0600.)

FIELD MANUALS

FM 3-34.214 - Explosives and Demolitions

(This Field Manual may be viewed and printed at http://www.apd.army.mil/, or ordered from Commandant, U.S. Army Engineer School, Ft. Leonard Wood, MO 65473-6650.)

US ARMY DEVELOPMENTAL TEST COMMAND

ITOP 1-2-601 - Laboratory Vibration Schedules,
ITOP 4-2-602 - Rough Handling

(Copies of these documents may be ordered from the US Army Developmental Test Command, ATTN: Publications, 314 Longs Corner Road, Aberdeen Proving Ground, MD 21005-5005, or online at http://www.dtc.army.mil/publications/topsindex.aspx.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design verification. When specified (see 6.2), a sample of the M7 shall be subjected to design verification in accordance with 4.1.

3.2 First article. When specified (see 6.2), a sample of the M7 shall be subjected to the first article inspection in accordance with 4.2.

3.3 Conformance inspection. When specified (see 6.2), a sample of the M7 shall be subjected to conformance inspection in accordance with 4.3.

3.4 Interface and interoperability.

3.4.1 Blasting cap body dimensions. The length of the blasting cap shall be not less than 2.25 inch and not greater than 2.35 inch. The outside diameter shall be not less than 0.230 inch and not greater than 0.241 inch.

3.4.2 Blasting cap body material. The blasting cap body shall be capable of protecting energetic material and sufficiently strong to allow insertion into M112 demolition blocks at room temperature. An extruded aluminum cup with side-wall thickness within the range of 0.008 to 0.011 inches has been found acceptable in the past.

3.4.3 Time fuse interface. The blasting cap shall be capable of being securely crimped to an M700 time fuse.

3.4.4 Energetic Material loading. Energetic materials shall be loaded to leave not less than 0.75 inches of open cap for safe insertion and crimping of time fuse.

3.4.5 Loose/Compacted Powder (critical). The M7 shall show no evidence of loose or compacted powder on the cup’s outer walls or inner walls above ignition column.

3.4.6 Cap split or cracked (critical). The blasting cap shall not be split or cracked to expose the explosive charge or show evidence of explosive leakage.

3.4.7 Inert blasting cap. Inert blasting cap shall utilize the same body as the energetic M7 except for loading and marking as indicated.

3.4.7.1 Inert blasting cap loading. M7 inert blasting cap shall use a non-energetic fill that contains no ferric iron oxide.

3.4.7.2 Inert blasting identification. M7 inert caps shall have four (4) holes drilled on the circumference of the inert blasting cap using a 35-43 drill bit size or its equivalent to

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

establish an unobstructed view through the inert blasting cap and inert fill. These holes shall be 90 degrees apart. The word “inert” is to be legible and marked on two sides opposite each other.

3.5 Operating requirements.

3.5.1 Initiation performance. The M7 shall be capable of being initiated by M700 Time Blasting Fuse. The blasting cap shall detonate completely as evidenced by initiating the following military explosives: C-4 blocks, 15/40 lbs. Shaped Charges, and TNT blocks.

3.5.2 Explosive output. The explosive output of the M7 shall be not less than 0.01 inches, measured in accordance with MIL-STD-331 Test D4.

3.5.3 Explosive ordnance disposal (EOD). The M7 shall be capable of being rendered safe and disposed of in all field environments and shall conform to the standard EOD render safe procedures. Such procedures for blasting caps typically include 30-minute wait time following malfunction; checking malfunction for low-order detonation or dud; and disposing of remaining material with new donor charge.

3.6 Support and ownership.

3.6.1 Reliability. The blasting cap shall have 99.9 percent functional reliability rating with 95% confidence.

3.6.2 Energetic material. The energetic material used in the blasting cap shall be qualified and approved for intended purposes by the Army Qualification Authority (see 6.9).

3.6.3 Energetic materials compatibility. All energetic materials (i.e. explosives, propellants, pyrotechnics, etc.) utilized shall be compatible with all combinations of directly contacting energetic and non-energetic materials. In addition, they shall be compatible with all combinations of materials which have the potential for contact in the system's life-cycle (i.e., manufacturing environment, storage, etc.).

3.6.4 Energetic sensitivity. All energetic materials shall be insensitive to initiation during routine shipping, handling and storage.

3.6.5 Human-system integration (HSI), AKA MANPRINT. The M7 shall be capable of being deployed and functioned safely and rapidly by a soldier used by military personnel wearing the environmental protective ensembles for cold weather gear and MOPP IV gear. These ensembles include at least the butyl hardware over the 5-finger wool insert while wearing the M40 gas mask and helmet protective cover (overgarment, undergarment, and vinyl overboots are not necessary.)

3.6.6 Ammunition lot numbering. Ammunition lot numbering shall be assigned in accordance with MIL-STD-1168 (see 6.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

3.6.7 Storage life. The M7 blasting cap in the packaged configuration shall remain safe and functional with no degradation in performance for not less than 20 years in protected storage and for not less than 2 years in unprotected storage in all climatic conditions.

3.6.8 Workmanship. All parts and assemblies shall be clean and free of burrs, sharp edges, unblended radii, cracks, gouges, surface defects, chips, dirt, grease, and oil (except where specifically required), corrosion, and other foreign matter. The cleaning method shall not be injurious to any of the parts or assembly nor shall they be contaminated by the cleaning agents.

3.6.9 Final hazard classification. When packaged in accordance with the packaging provisions of the contract, the blasting caps shall have no more severe hazard classification than the following:

Single cap configuration:	DOD Hazard Class/Div: 1.4 DOD Hazard Compatibility Group: B DOT Hazard Class: 0255 UN Number: 0255 UN NOMENCLATURE: DETONATORS, ELECTRIC

3.7 Environmental requirements. The M7 shall function after exposure to the following environmental conditions.

3.7.1 Operating temperatures. The M7 shall show no evidence of damage or failure and shall function throughout the temperature range of -50°F to 125°F.

3.7.2 Transit and storage temperature. The M7 shall show no evidence of damage or failure and shall function at the operating temperatures after exposure to storage temperatures from -65ºF to +165ºF.

3.7.3 Temperature shock. The M7 shall show no evidence of damage or failure and shall function after exposure to thermal shock.

3.7.4 Humidity. The M7 in its inner package (M19A1 ammunition can or as specified by contract) shall show no evidence of damage or failure and shall function after exposure to high temperature of $140^{o}F + 5^{o}F$, and a relative humidity of 95 percent.

3.7.5 Solar radiation. The M7 in its inner package (M19A1 ammunition can or as specified by contract) shall show no evidence of damage or failure and shall function after exposure to solar radiation.

3.7.6 Fungus. The M7 in its inner package (M19A1 ammunition can or as specified by contract) shall show no evidence of damage or failure and shall function after exposure to insects, fungi, and molds.

3.7.7 Salt/fog. The M7 in its inner package (M19A1 ammunition can or as specified by contract) shall show no evidence of damage or failure and shall function after being subjected to the effects of salt/fog.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

3.7.8 Dust. The M7 in its inner package (M19A1 ammunition can or as specified by contract) shall show no evidence of damage or failure and shall function after exposure to the effects of dust.

3.7.9 Sand. The M7 in its inner package (M19A1 ammunition can or as specified by contract) shall show no evidence of damage or failure and shall function after exposure to the effects of sand.

3.7.10 Icing/freezing rain. The M7 in its inner package (M19A1 ammunition can or as specified by contract) shall show no evidence of damage or failure and shall function after exposure to icing/freezing rain.

3.7.11 Lightning. The M7 in its inner package (M19A1 ammunition can or as specified by contract) shall show no evidence of damage or failure and shall not function as a result of a near lightning strike in any configuration from stockpile to mission completion.

3.7.12 Six-foot drop. The M7 blasting cap shall remain safe and fully functional after exposure to two consecutive six-foot drops onto a surface of ¾ inch plywood securely clamped to a steel anvil of not less than 3 inches thickness..

3.7.13 Transportation/vibration. The M7 in its inner package (M19A1 ammunition can or as specified by contract) shall not exhibit any evidence of damage or failure when subjected to transportation/vibration.

3.7.14 Stacking. The M7 in its fully packaged configuration (outer pack) shall be rugged enough to withstand a stacking load equal to the weight of stacking like outer packs, including pallets to a height of 16 feet.

3.7.15 Rough handling. The M7 in its fully packaged configuration (outer pack) shall not exhibit evidence of damage or failure after undergoing a sequence of rough handling and transportation vibration.

4. VERIFICATION

TABLE I: Requirement/verification cross reference matrix.

METHOD OF VERIFICATION	**CLASS OF VERIFICATION**	**Conformance Verification**
1 - ANALYSIS	A - DESIGN VERIFICATION	I, II: Inspection Levels of Table 4
2 - DEMONSTRATION	B - FIRST ARTICLE	a. See Table V
3 - EXAMINATION	C - CONFORMANCE	b. See Note 1
4 - TEST		c. See Note 2

Section 3	Verification Method				Verification Class			Section 4
	1	2	3	4	A	B	C	
3.4 Interface and interoperability								
3.4.1 Blasting cap body			X		X	X	VL-IV	4.4.1
3.4.2 Blasting cap materials			X		X	X	VL-IV	4.4.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

TABLE I: Requirement/verification cross reference matrix – continued.

Section 3	Verification Method				Verification Class			Section 4
	1	2	3	4	A	B	C	
3.4.3 Time fuse interface			X		X	X	VL-IV	4.4.3
3.4.4 Energetic material loading		X	X	X	X	X	VL-IV	4.4.4
3.4.5 Loose/compacted powder			X		X	X	100%	4.4.5
3.4.6 Cap split or cracked			X		X	X	100%	4.4.6
3.4.7 Inert blasting cap			X			X	VL-IV	4.4.7
3.4.7.1 Inert blasting cap loading			X		X	X	VL-IV	4.4.7.1
3.4.7.2 Inert blasting ident.			X		X	X	VL-IV	4.4.7.2
3.5 Operating requirements								
3.5.1 Initiation performance		X			X	X		4.5.1
3.5.2 Explosive output		X			X	X	VL-IV	4.5.2
3.5.3 Explosive ord. disposal		X			X			4.5.3
3.6 Support and ownership								
3.6.1 Reliability			X		X			4.6.1
3.6.2 Energetic material	X				X			4.6.2
3.6.3 Energetic materials compatibility	X				X			4.6.3
3.6.4 Energetic sensitivity		X	X	X	X			4.6.4
3.6.5 MANPRINT	X	X			X			4.6.5
3.6.6 Ammo. lot numbering			X	X			VL-II	4.6.6
3.6.7 Storage life	X	X	X	X	X			4.6.7
3.6.8 Workmanship			X	X	X	X	VL-II	4.6.8
3.6.9 Final hazard classification		X	X		X			4.6.9
3.6.10 Insensitive munitions		X	X		X			4.6.10
3.7 Environmental requirements								
3.7.1 Operating temperatures		X	X		X	X		4.7.1
3.7.2 Transit and storage temp.		X	X		X			4.7.2
3.7.3 Temperature shock		X	X		X			4.7.3
3.7.4 Humidity		X	X		X			4.7.4
3.7.5 Solar radiation		X	X		X			4.7.5
3.7.6 Fungus		X	X		X			4.7.6
3.7.7 Salt/fog		X	X		X			4.7.7
3.7.8 Dust		X	X		X			4.7.8
3.7.9 Sand		X	X		X			4.7.9
3.7.10 Icing/freezing rain		X	X		X			4.7.10
3.7.11 Lightning		X			X			4.7.11
3.7.12 Six-foot drop				X	X	X	VL-IV	4.7.12
3.7.13 Transportation/vibration		X	X		X			4.7.13
3.7.14 Stacking		X	X		X			4.7.14
3.7.15 Rough handling		X	X		X			4.7.15

4.1 Design verification. When specified, a sample of the M7 shall be subjected to design verification in accordance with TABLES I and II.

4.1.1 Design verification test rejection. If any test sample fails to comply with the design verification requirements, the design shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

TABLE II. Design verification test sequence and quantity.

Requirement	Qty	Test Method
3.4.1	100	4.4.1 Use same samples for 3.4.1 thru 3.4.4.
3.4.2	100	4.4.2
3.4.3	100	4.4.3
3.4.4	100	4.4.4
3.4.5	All samples	4.4.5
3.4.6	All samples	4.4.6
3.5.1	20	4.5.1
3.5.2	20	4.5.2
3.5.3	22	4.5.3
3.6.1	3000 total	4.6.1 Perform following all function testing.
3.6.2	Quantities as required	4.6.2
3.6.3	Quantities as required	4.6.3
3.6.4	Quantities as required	4.6.4
3.6.5	20	4.6.5
3.6.7	40	4.6.7
3.6.8	60	4.6.8
3.6.9	15 outer packs	4.6.9
3.6.10	120	4.6.10
3.7.1	1200	4.7.1 See Note.
3.7.2	600	4.7.2
3.7.3	40	4.7.3
3.7.4	1 inner pack	4.7.4
3.7.5	1 inner pack	4.7.5
3.7.6	1 inner pack	4.7.6
3.7.7	1 inner pack	4.7.7
3.7.8	1 inner pack	4.7.8
3.7.9	1 inner pack	4.7.9
3.7.10	1 inner pack	4.7.10
3.7.11	1 inner pack	4.7.11
3.7.12	400	4.7.12.1
3.7.13	1 inner pack	4.7.13
3.7.14	1 outer pack	4.7.14
3.7.15	6 outer packs	4.7.15
Note: Test equal numbers of caps at each of the specified temperatures.		

4.2 First article inspection. When specified, a sample of the M7 shall be subjected to first article verification test in accordance with TABLES I and III.

4.2.1 First article rejection. If any test sample fails to comply with the first article requirements, the samples shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

TABLE III. First article inspection sequence and quantity.

Requirement	Qty	Test Method
3.4.1	25	4.4.1
3.4.2	25	4.4.2
3.4.3	25	4.4.3
3.4.4	25	4.4.4
3.4.5	All samples	4.4.5
3.4.6	All samples	4.4.6
3.4.7	10	4.4.7
3.4.7.1	10	4.4.7.1
3.4.7.2	10	4.4.7.2
3.5.1	5	4.5.1
3.5.2	10	4.5.2
3.6.8	25	4.6.8
3.7.1	30	4.7.1 See Note.
3.7.12	125	4.7.12
Note: 10 at cold temperature requirement and 20 at hot temperature requirement.		

4.3 Conformance verification.

4.3.1 Conformance verification. When specified, a sample of the M7 blasting caps shall be subjected to conformance verification in accordance with TABLE I and IV.

TABLE IV. Conformance inspection sequence and quantity.

Requirement	Qty	Test Method
3.4.1	VL-IV	4.4.1
3.4.2	VL-IV	4.4.2
3.4.3	VL-IV	4.4.3
3.4.4	VL-IV	4.4.4
3.4.5	100%	4.4.5
3.4.6	100%	4.4.6
3.4.7	VL-IV	4.4.7
3.4.7.1	VL-IV	4.4.7.1
3.4.7.2	VL-IV	4.4.7.2
3.5.2	VL-IV	4.6.1, 4.6.2.1
3.6.6	VL-II	4.6.6
3.6.8	VL-II	4.6.8
3.7.12	VL-IV	4.7.12

4.3.2 Classification of characteristics. Conformance inspection shall include the examination and tests listed in Table I using the applicable attribute sampling plan. All verifications are classified as major characteristics except for 4.6.6 and 4.6.8 which are classified as minor characteristics. 4.4.5 and 4.4.6 are critical characteristics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

4.3.3 Conformance inspection rejection. If any test sample fails to comply with the applicable acceptance criteria (See Tables I, IV and V) the lot from which the sample were taken shall be rejected.

4.4 Interface and interoperability verification.

4.4.1 Blasting cap body. The outside diameter and the length of the blasting cap (3.4.1.2.1) shall be measured using standard measuring equipment capable of measuring down to a tenth of a thousandth of an inch and capable of measuring down to one thousandth of an inch, respectively.

4.4.2 Blasting cap material. Verify by demonstration that material selected in the design of the M7 is sufficiently strong to allow insertion into designated demolition materials (see 3.5.1).

4.4.3 Time fuse interface. Verify secure crimping by affixing the blasting cap horizontally to a fixture and apply a minimum 5 lbf pull force on the time fuse.

4.4.4 Energetic Material Loading. Verify by examination or analysis of Statistical Process Control data that energetic materials have been pressed to leave sufficient void at the open end of the cap.

4.4.5 Loose/Compacted Powder (critical). The M7 shall be inspected for loose/compacted powder on the outer wall of the cup and inner wall above the ignition column.

4.4.6. Cap split or cracked (critical). The M7 shall be visually inspected for splits or cracks in the cap.

4.4.7 Inert blasting cap. Verify by examination that inert caps use same body as energetic caps.

4.4.7.1 Inert blasting cap loading. Verify by examination of objective evidence presence of filler.

4.4.7.2 Inert blasting identification. Verify by examination the location of drilled holes and presence of “inert” marking.

4.5 Operating requirement verification.

4.5.1 Initiation performance verification. Verify by test that the M7 blasting cap functions high-order the following demolition items: two 15 lbs/40 lbs Shaped Charges, two M030 TNT demolition blocks, and the remainder of test requirement with M112 C-4 demolition blocks. The demolition items shall be conditioned to not greater than -35ºF and not less than -40ºF. Priming of the demolition items shall be in accordance with FM 3-34.214.

4.5.2. Explosive output verification. The output of the M7 shall be measured in accordance with MIL-STD-331 Test D4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

4.5.3 Verification of Explosive Ordnance Disposal (EOD). When specified, a 22 sample quantity of M7 shall be subjected to EOD verification by the Government.

4.6 Support and ownership verification.

4.6.1 Reliability. The reliability shall be determined by analysis following the completion of all function testing.

4.6.2 Energetic material qualification. Objective evidence of Army Qualification Authority approval shall be examined (see 6.9).

4.6.3 Energetic material compatibility. Compatibility tests shall be conducted in accordance with Method 504.1.1 of MIL-STD-650 or Method 408.1.1 of MIL-STD-286C. Compatibility is demonstrated when results reflect negligible reactivity.

4.6.4 Energetic sensitivity. Energetic sensitivity test shall be conducted in accordance with the test procedures listed below. Acceptable sensitivity is demonstrated when results comply with the requirements listed in TABLE VI.

TABLE VI. Energetic sensitivity tests.

Test	Reference Document	Requirement
ERL/Bruceton	4.7.3.1 (AOP-7 (Edition 2) US 201.01.001)	50% Initiation @ 20 cm (min.)
Friction Steel/Fiber Shoe	4.7.3.2 (AOP-7 (Edition 2) US 201.02.008)	Burn with no detonation (max. reaction)
Electrostatic Discharge @ 0.030 Joule	4.7.1.1 (AOP-7 (Edition 2) US 201.03.001)	0 reactions in 20 tests
Differential Thermal Analysis	4.7.3.3 (AOP-7 (Edition 2) US 202.01.007)	200 °C (min.)

4.6.5 Human system integration. Verify by demonstration that the M7 can be used by military personnel wearing the environmental protective ensembles as specified in 3.6.5. The samples shall function as per paragraph 4.5.2.

4.6.6 Ammunition lot numbering. Verify that an ammunition lot numbering has been assigned to each lot of the M7 as per MIL-STD-1168.

4.6.7 Storage life verification. Verify the M7 blasting caps meet the storage life requirement by conducting the MIL-STD-331 test C1. At the completion of test, blasting cap assemblies shall be function tested per 4.5.1 – 4.5.2.

4.6.8 Workmanship verification. Verify the workmanship on the production of the M7 blasting cap by visual inspection. The lot shall be inspected for any burrs, sharp edges, unblended radii, gouges, surface defects, corrosion, chips, dirt, grit, grease, or other foreign material. Evidence of said impurities shall be cause for lot rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

4.6.9 Final hazard classification. Compliance with the FHC requirements specified of paragraph 3.6.9 shall be validated during the Performance Verification Test (PVT). PVT tests shall be in accordance with TB 700-2, Department of Defense Ammunition and Explosives Hazard Classification Procedures. The following UN test series, as defined in TB 700-2, shall be used: 6(a) for single package test, 6(b) for stack test, 6(c) for external fire test, 4(a) for thermal stability and 4(b) (ii) for 12 meter drop test.

4.6.10 Insensitive munition certification. The M7 must be subjected to the Sympathetic Detonation test in accordance with MIL-STD-2105. One donor outerpack and two acceptor outerpacks shall be used for this test. Simulated (dummy) outerpacks shall also be used to simulate the condition of the M7 packaged munition items. The passing criterion for this test is that when the donor M7 is initiated no sympathetic detonation or burning is allowed in the donor outerpack nor in the acceptor outerpacks.

4.7 Environmental verification.

4.7.1 Operational temperatures.

a. Hot temperature. Condition the M7 blasting cap at not less than 120ºF and not greater than 125ºF for a minimum of 24 hours. Perform functional test in accordance with procedure in 4.5.2 within five (5) minutes after the samples have been removed from the conditioning chamber.

b. Ambient temperature. Condition the M7 blasting cap at not less than 75ºF and not greater than 85º F for a minimum of 24 hours. Perform functional test in accordance with procedure in 4.5.2 within five (5) minutes after the samples have been removed from the conditioning chamber.

c. Cold temperature. Condition the M7 blasting cap at not greater than -45º F and not less than -50º F for a minimum of 24 hours. Perform functional test in accordance with procedure in 4.5.2 within (5) minutes after the samples have been removed from the conditioning chamber.

4.7.2 Transit/storage temperatures. Two groups of equal number hot and cold of the M7 shall be conditioned for 72 hours, minimum. Prior to inspection, samples shall be stabilized in the range of 75º F to 85º F. Within 15 minutes of removal of the samples, examine each sample for compliance and then initiate.

4.7.3 Temperature shock test. Perform the following test on M7 for a minimum of 3 cycles between the temperature extremes of paragraph 3.9.3 in accordance with MIL-STD-810, Method 503.4, Procedure II. The minimum duration of the low and the high temperatures is (1) one hour. Within 15 minutes after the completion of the conditioning, examine the M7 for damage and then initiate.

4.7.4 Humidity. Perform the test for five (5) cycles in accordance with MIL-STD-810, Method 507.4, with the chamber temperature variation as per figure 507.4-1. Within 15 minutes after completion of the conditioning, examine the M7 and then initiate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

4.7.5 Solar radiation test. Perform the solar radiation test as per MIL-STD-810, Method 505.3, Procedure II for ten (10) 24 hour cycles on the M7. Within 15 minutes of completion of this conditioning, inspect the individual M7's for signs of damages per MIL-STD-810, Method 505.3, Paragraph I-4.1. Any M7 sustaining such damages shall be considered as a failure. The conditioned samples shall be initiated.

4.7.6 Fungus test. Perform the test for a duration of 84 days (see paragraph I-3.2a) in accordance with MIL-STD-810, Method 508.4, Procedure II. Inspection shall be in accordance with MIL-STD-810, Method 508.4, Paragraph I-4.1. Within 15 minutes from removal from the conditioning chamber, inspect and initiate the M7.

4.7.7 Salt/fog test. Perform a 96 hour, i.e. 4-24 hour cycles, salt spray of a 5% concentration (fog) of the M7 in accordance with MIL-STD-810, Method 509.4. Upon completion of this test procedure, inspect the M7 for evidence of surface degradation such as flaking, pitting, blistering or separation of finish. Within 15 minutes from the completion of test and the removal of the samples from the conditioning chamber, inspect and initiate the M7.

4.7.8 Dust test. Perform the blowing dust test on the M7 per MIL-STD-810, Method 510.4, Procedure I (air velocity of 8.9m/sec). Inspection shall be in accordance with MIL-STD-810, Method 510.4, Paragraph I-4.1. Within 15 minutes from removal from the conditioning chamber, inspect and initiate the M7.

4.7.9 Sand test. The M7 shall be subjected to the blowing sand test in accordance with MIL-STD-810, Method 510.4, paragraph I.3.2d.(1)(b), Procedure II (air velocity of 29m/sec.). Inspection shall be in accordance with MIL-STD-810, Method 510.4, Paragraph I-4.1. Within 15 minutes from removal from the conditioning chamber, inspect and initiate the M7.

4.7.10 Ice and freezing rain test. The M7 shall be subjected to the ice and freezing rain test in accordance with MIL-STD-810, Method 521.1 at temperature -35 - 15°F to establish a uniform glaze ice coating. Inspect the M7 for evidence of damage. Within 15 minutes from removal from the conditioning chamber, inspect and initiate the M7.

4.7.11 Lightning effect test. The M7 shall be exposed to near lightning strike at 10 meters, as defined in MIL-STD-464 with the following characteristics: 200,000 ampere peak, time to peak 1.5 microseconds, time to half current 50 microseconds, 400 amperes continuing current last 400 milliseconds.

4.7.12 Six-foot drop. The bare M7 shall be dropped from a height of six feet onto its open end using a test stand to ensure proper orientation of the cap. Tape shall cover the open end of the cap to aid in identifying whether explosive powder was knocked loose. Any loose powder or other irregularity shall be deemed a failure. Dropped caps shall be initiated as specified in 4.5.2.

4.7.13 Transportation/vibration. The ammunition can configuration (inner pack) of the M7 shall be subjected to ITOP 1-2-601. Verify that each item does not exhibit any evidence of damage or failure. After inspection, the item shall be initiated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

4.7.14 Stacking. A minimum of one (1) outer pack shall be used for this test. The load to be imposed on top of the outer pack shall be calculated based on the packed out weight of the individual outer pack multiplied by number of outer packs that could be stacked to a height of 16 feet. Additional weight shall be included to represent the additional load imposed as a result of the pallet weight. The test shall be conducted at ambient temperature for a minimum of 24 hours. The outer pack shall exhibit no signs of distortion, deformation or cracks after exposure under load for the stated time.

4.7.15 Rough handling. Two groups of six (6) outer packs each of M7 shall be exposed to the following tests after one group is conditioned at +160°F + 5°F for a minimum of two (2) hours and the other is conditioned at -60°F - 5°F for a minimum of two (2) hours. NOTE: Units Under Test are removed from the outer packs after the completion of the following four (4) tests.

a. Test 1: Perform a Secured Cargo Vibration test in accordance with ITOP 1-2-601.

b. Test 2: Perform six (6) consecutive 3 Foot Drops per pack in the following orientation:

(1) Bottom
(2) Top
(3) Side
(4) Left End
(5) Bottom Left Corner at 45°
(6) Bottom Right End Edge (Corner) at 45°

Conduct this test on the quantity of packs used in the secured cargo vibration test. The impact surface shall be as defined in ITOP 4-2-602.

NOTE: No more than three (3) minutes shall lapse from the time when a pack is removed from the conditioning chamber to when the first drop is performed. No more than ten (10) minutes shall lapse between removal from the conditioning chamber to the completion of the six (6) drops on each container.

c. Test 3. Perform a loose cargo vibration test in accordance with ITOP 4-2-602. Conduct this test on the quantity of packs used in the 3 foot drop test (Test 2).

d. Test 4: Perform a Seven Foot Drop test in accordance with ITOP 4-2-602. Conduct this test on the quantity of packs used in the loose cargo vibration test (Test 3). Conduct one (1) seven foot drop per pack. Each box shall be dropped in a different orientation. The orientations are as described above under the 3 Foot Drop test (Test 2). Verify compliance with paragraph 3.7.15. Randomly select 7 M7's from each of the 4 inner packs that were randomly selected from the 6 outer packs. Within 15 minutes of the completion of test 4, examine and initiate in accordance with paragraph 4.5.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The M7 covered by this specification, are intended for use by military forces to not only initiate demolition items in general but in particular to initiate a multiple number of items. These blasting caps, covered by this specification are military unique as they must function under rough handling and extreme climatic and storage conditions as encountered by the United States Armed Forces.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification and of all documents cited in Section 2 and listed in Section 6 for information.
b. Packaging requirements (see 5.1).
c. Requirements for design verification (see 4.2).
d. Requirements for first article (see 4.3).
e. Special packaging requirements apply. Design and test requirements for level A ammunition packaging should be in accordance with MIL-STD-1904 (Acceptance Criteria: Ammunition).
f. Performance oriented packaging requirements apply. Contact U.S. Army ARDEC, ATTN: AMSRD-AAR-AIL-P, Picatinny, New Jersey 07806-5000.
g. Markings should be in accordance with drawing 12982865.
h. Serialized requirements, if applicable.
i. Certificate of conformance for each lot or shipment of product must be obtained.
j. Government furnished equipment (see 3.4) (if applicable).
k. Information required to satisfy the IM technical performance criteria of MIL-STD-2105B, and IM programmatic requirements of DA Pamphlet 70-3. Acquisition procedures contained in DA Pamphlet 70-3, Appendix 25 will be followed (see 3.6.10).
l. If the final hazard classification on record was based on government packaging drawings it may be necessary to include those packaging drawings in the contract to insure continued legal transportation. Determination should be made by AMSRD-AAR-QEP-B.
m. Requirements for acceptance inspection equipment (AIE) designs, (see 6.3).
n. Ammunition Data Cards and Ammunition Lot Numbering Should be IAW MIL-STD-1168.(see 6.4).
o. Special compatibility, and energetic material sensitivity data applies. Contact AMSRD-AAR-QES-B for requirements.
p. Requirements for EOD qualification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

q. Requirements for Critical Safety Inspection. A note must be included in the contract to advise that experience has shown that 100 percent visual and 100 percent manual inspections are usually unsatisfactory for this purpose.

r. Applicable national stock number (NSN).

s. Acquisition documents must specify the following: Information needed to satisfy the requirements for qualification of energetic materials by the Army Service Qualification Authority, see 6.8. This information can be obtained in the desktop guide titled Explosive Hazard Classification Requirements for Performance Specifications that is available from AMSRD-AAR-QEP-B, Bldg. 62, Picatinny Arsenal, NJ 07806-5000.

t. Requirements for qualification of EOD (see 6.5).

6.3 Acceptance inspection equipment (AIE). Provisions concerning the AIE and the associated procedures used to verify the requirements of this specification should be specified in the contract.

6.4 Performance oriented packaging (POP). The inner and exterior packs cited above should meet all the POP test requirements in accordance with Code of Federal Regulations, Title 49 (49CFR), including testing and re-testing as specified in Paragraph 178.601(e) of 49CFR. A POP test report should be generated in accordance with DI-PACK-81059 following the tests. When completed the POP test report must be kept on file by the contractor and must also be submitted to the U.S. Army Armament Research Development and Engineering Center, ATTN: AMSRD-AAR-AIL-P, Picatinny, New Jersey, 07806-5000. The POP marking symbol applied to the exterior pack should be that belonging to the organization which conducted the POP testing.

6.5 Ammunition lot numbers. Ammunition lot numbers require ammunition data cards in accordance with MIL-STD-1168.

6.6 EOD qualification. Qualification for EOD by the Army requires demonstration that the design is capable of being rendered safe and disposed of in all field environments. This task will require samples to be tested to EOD procedures, design information and test data. Point of contact for further information on EOD qualification can be obtained from U.S. Army ARDEC, ATTN: AMSRD-AAR-AEX, Picatinny, NJ 07806-5000.

6.7 Government furnished property. Any government furnished property will be identified in the solicitation.

6.8 Submission of alternative conformance provisions. Proposed alternative conformance provisions will be submitted to the Government for evaluation/approval.

6.9 Energetic material qualification information. Qualification of energetic materials by the Army Service Qualification Authority requires demonstration that energetic materials meet the requirements of STANAG 4170. This task is addressed in the statement of work in the contract. Points of Contact for the Army Service Qualification Authority can be obtained from AMSRD-AAR-QES-C, U.S. Army ARDEC, ATTN: Reliability Management Office, AMSRD-AAR-QES, bldg 62, Picatinny Arsenal, NJ 07806-5000.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.

6.10 Waiving environmental requirements. When standard packaging materials and configurations are used, and with the Government's approval, the following Environmental requirements may be waived from Design Verification Testing: Humidity, Solar radiation, Fungus, Salt/fog, Dust, Sand, Icing/freezing rain, and Lightning. In order to meet the Reliability requirement an additional 280 items should be functioned in accordance with 4.5.2.

6.10 Subject term (key word) listing.

Cap, Blasting

Custodian:
Army-AR

Preparing Activity:
Army-AR
(Project: 1375-2008-002)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T05:43Z
Check the source to verify that this is the current version before use.